

Coding for Cybersecurity — Languages to Focus On

2026 Edition • SOC / Blue Team first, Pentest-ready • No software engineering degree needed

The rule

In cyber you need to be a good scripter and a good reader of other people's code. Automate tasks, parse logs, understand how an app breaks — not build full apps.

1. Python — learn this first

- Your workhorse for: log parsing, scanners, API calls, exploit PoCs, SIEM scripts.
- Learn: variables, loops, if/else, functions, files, requests/http, sockets TCP client, regex, argparse, json/csv, BeautifulSoup.
- Skip for now: Django/Flask, deep OOP, data science libraries.
- Cyber projects: port scanner, log IOC parser, hash checker.

2. Bash / Linux Shell — non-negotiable

- You will live in the terminal. Chain tools, triage hosts, quick one-liners.
- Learn: grep, awk, sed, find, xargs, pipes, redirection, for loops, cron.
- This speeds up 70% of SOC analyst daily work.

3. SQL — read and break databases

- For investigating DB logs, SQL injection, querying SIEM backends.
- Learn: SELECT, WHERE, JOIN, UNION, GROUP BY.
- Then: ' OR 1=1 -- style injections, and parameterized queries as the fix.
- You do not need DBA level.

4. JavaScript + HTML — for Web App Security

- Needed for bug bounty / pentest / AppSec. Skip initially if pure SOC/Blue Team.
- Learn: forms/cookies/sessions, DOM, fetch/XHR, XSS in browser console, JSON.
- You do not need React/Node.

5. PowerShell — for Windows / SOC / IR

- Most Indian enterprise endpoints are Windows.
- Learn: Get-Process, Get-EventLog, Get-NetTCPConnection, pipelines.
- Know how to read attacker PowerShell in logs: obfuscation, base64, Invoke-Expression.
- Learn this after Python + Bash, right before SOC interviews.

What to safely ignore at the start

C/C++, Java, Go, Rust, mobile languages, full-stack frameworks. Useful later for exploit dev / malware RE / secure code review, but they will slow you down now. Come back to C once Python is comfortable.

6-Week Coding-for-Cyber Plan — 1.5 hrs/day

Week	Focus	Daily exercise	End-of-week deliverable
1	Bash / Linux Shell	Day 1-2: navigation, grep/find. Day 3-4: pipes, awk/sed. Day 5: Headers, one-liner, log to find failed SSH logins, top talk	Day 5 Headers, one-liner, log to find failed SSH logins, top talk
2-3	Python Basics	D1: syntax/loops/files. D2: requests + json. D3: regex. D4: sockets TCP client. D5: save as CSV. D6-7: CLI. D6-7: IP	IP finder, sockets TCP client, save as CSV, CLI. D6-7: IP
4	Python for Security	D1-2: log parser for auth.log — count failed IPs. D3: hash checker, sys+VirusTotal API. D4-5: subdomain brute via	log parser for auth.log, hash checker, sys+VirusTotal API, subdomain brute via
5	SQL + Web JS	D1-2: SQL SELECT/JOIN/UNION. D3-4: PortSwigger SQLmap. D5: finding MITM for CS/cookies. D6: JS DOM XSS in	SQLmap, finding MITM for CS/cookies, JS DOM XSS in
6	PowerShell + Final	D1-2: Get-Process, Get-EventLog, Get-NetTCPConnection. D3: Get-NetBase64. D4: PowerShell logs. D5: build 3-pro	Get-NetBase64, PowerShell logs, build 3-pro

Command Cheat Sheet

Lang	Quick commands
Python	requests.get(url), open('file').read(), re.findall(r'\d+\.\d+\.\d+\.\d+', text), socket.socket(), json.load(), argparse
Bash	grep -Rin 'failed' /var/log, find / -perm -4000 2>/dev/null, awk '{print \$1}' sort uniq -c sort -nr, netstat -tulpn
SQL	SELECT user, COUNT(*) FROM logins WHERE status='failed' GROUP BY user HAVING COUNT(*) > 5;
JS	document.cookie, fetch(url), JSON.parse(), <script>alert(1)</script> test
PowerShell	Get-EventLog Security -InstanceId 4625, Get-NetTCPConnection, [Convert]::FromBase64String()

Portfolio — 3 scripts that get interviews

- 1. port_scanner.py — Python TCP scanner, CSV output
- 2. log_parser.py — parses auth.log, top 10 attacker IPs, JSON output
- 3. hash_checker.py — checks file SHA256 against a local IOC list
- Add a README for each: what it does, how to run, sample output, what you learned.

Study rhythm

45 min concept → 45 min hands-on coding → 15 min notes. Close laptop, write what you remember. Green = can teach it, Yellow = needs notes, Red = redo lab.

Generated June 2026 — Coding for Cybersecurity, SOC-entry track. Practice only on systems you own or have permission for.